

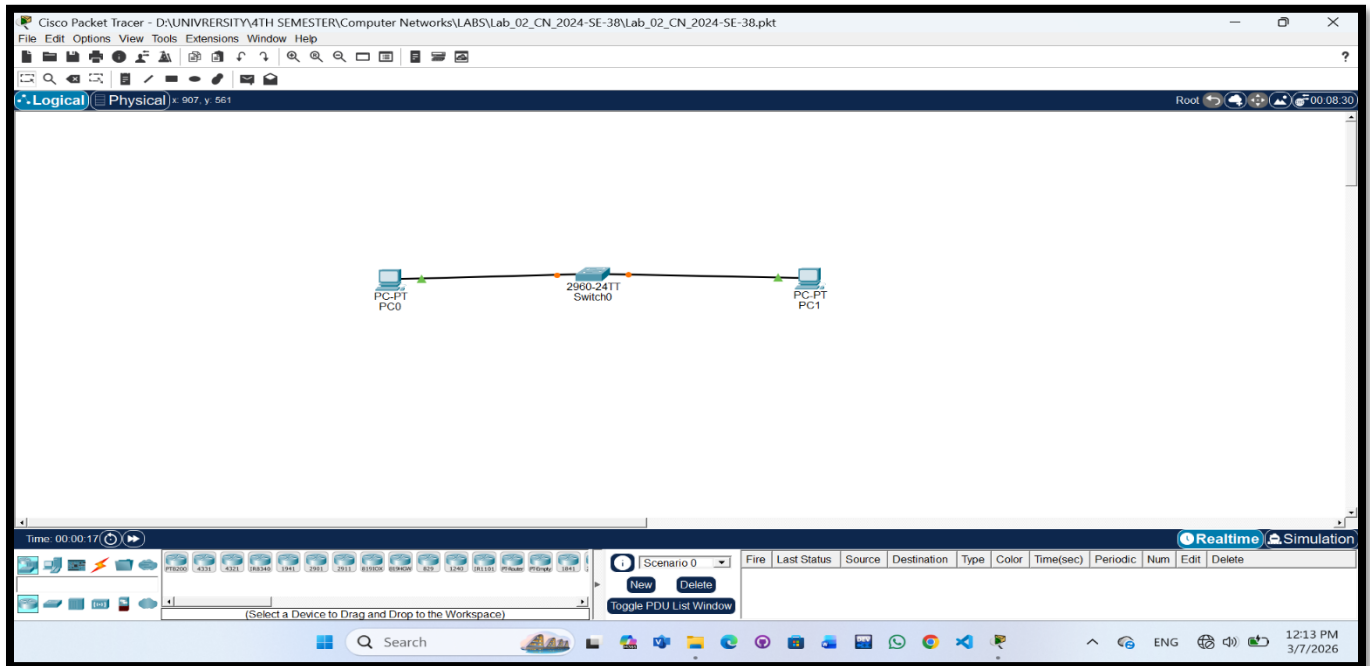


The University of Azad Jammu and Kashmir, Muzaffarabad

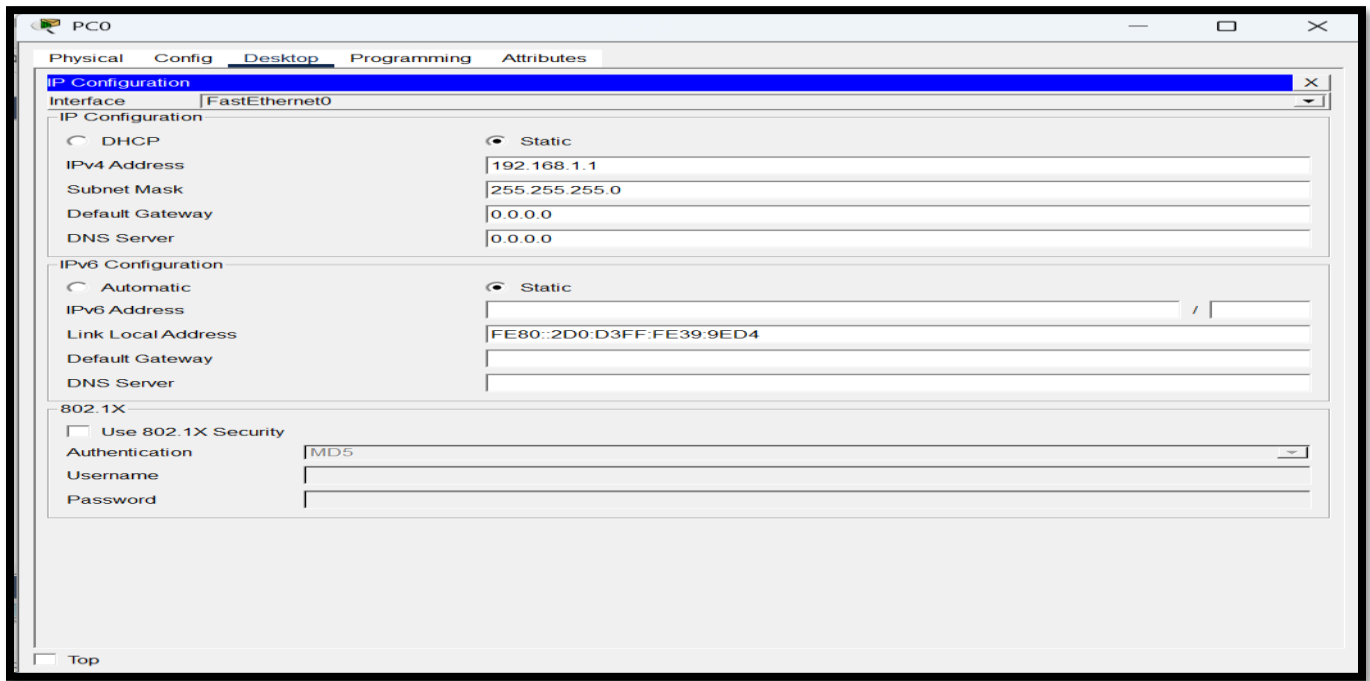
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Submitted to	Engr. Dr. Asma Jayaid
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Session	2024-2028
Roll No	2024-SE-38
Lab No	02
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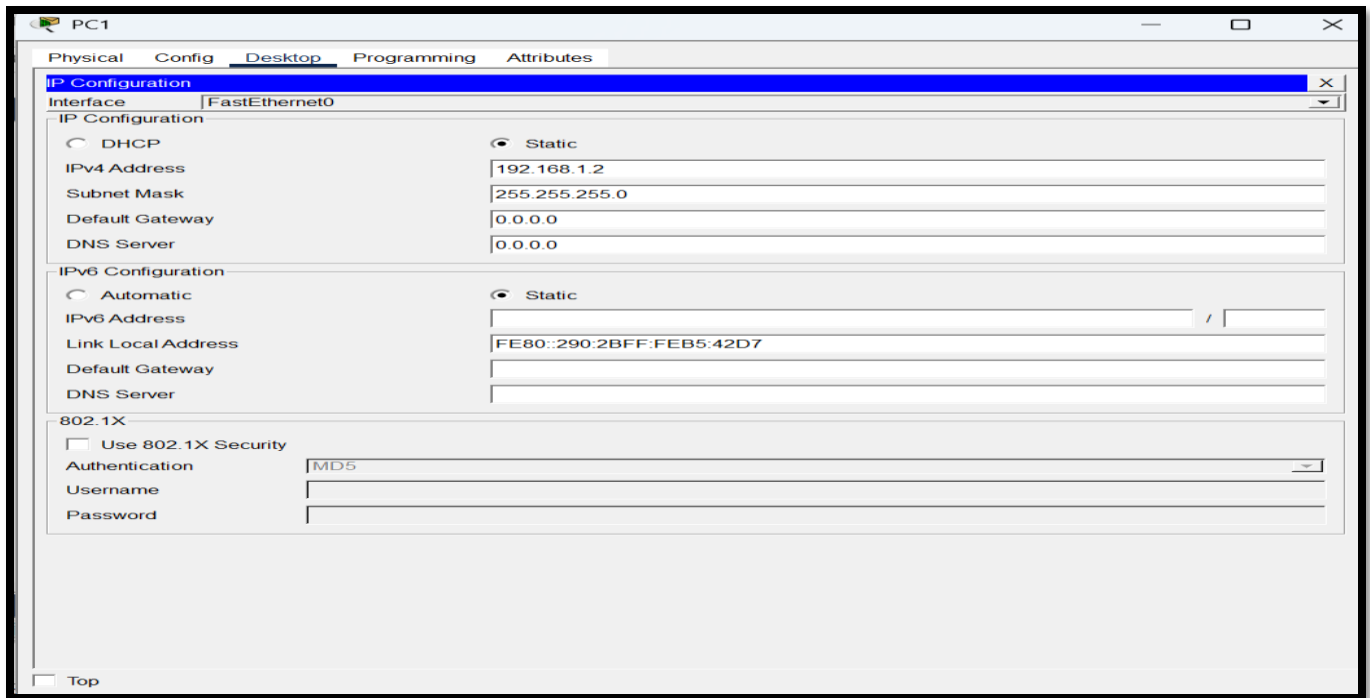
Create and Test a Simple LAN in Cisco Packet Tracer

Step 01: *Open Cisco Packet Tracer and take one switch and two PCs.*



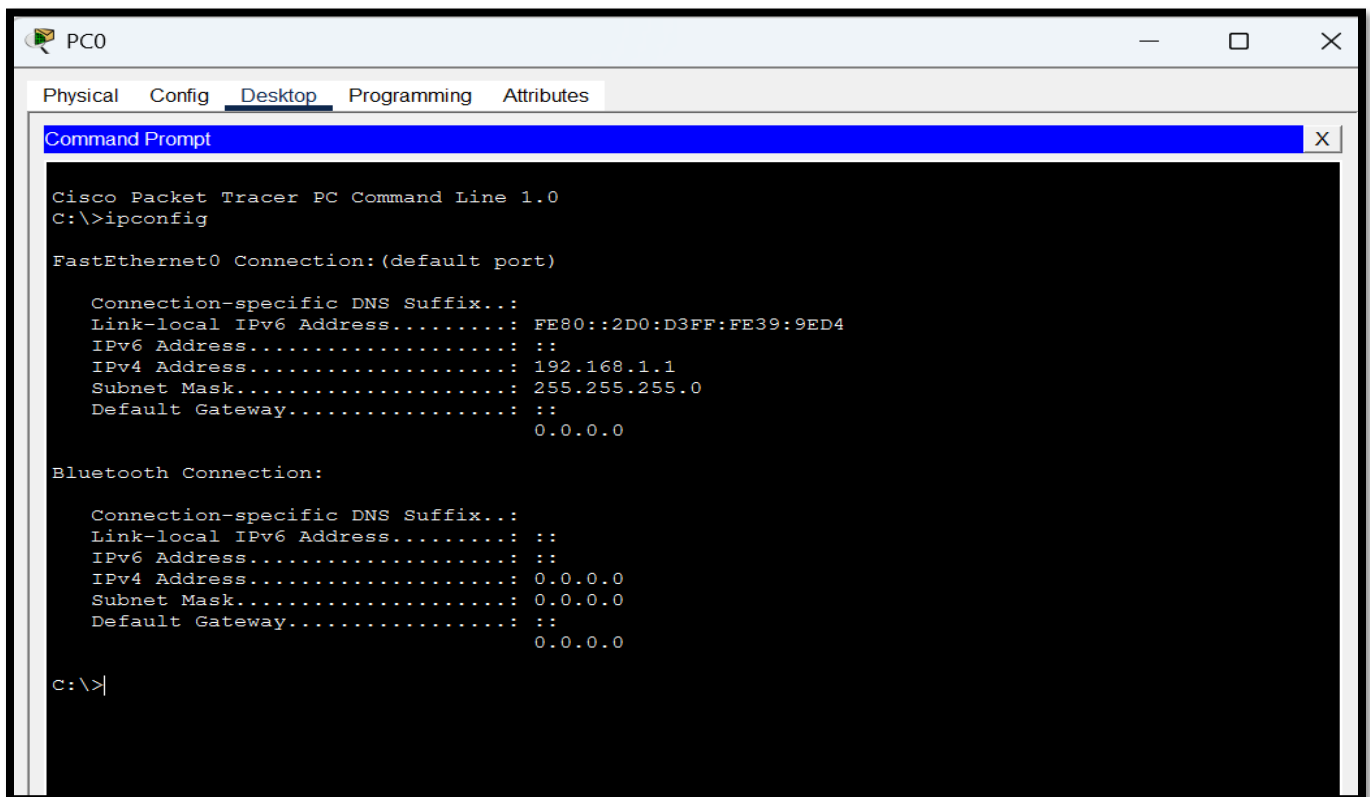
Step 02: *Click on the PC. Go to the Desktop tab. Click on IP Configuration. Enter the IP Address and Subnet Mask as instructed.*





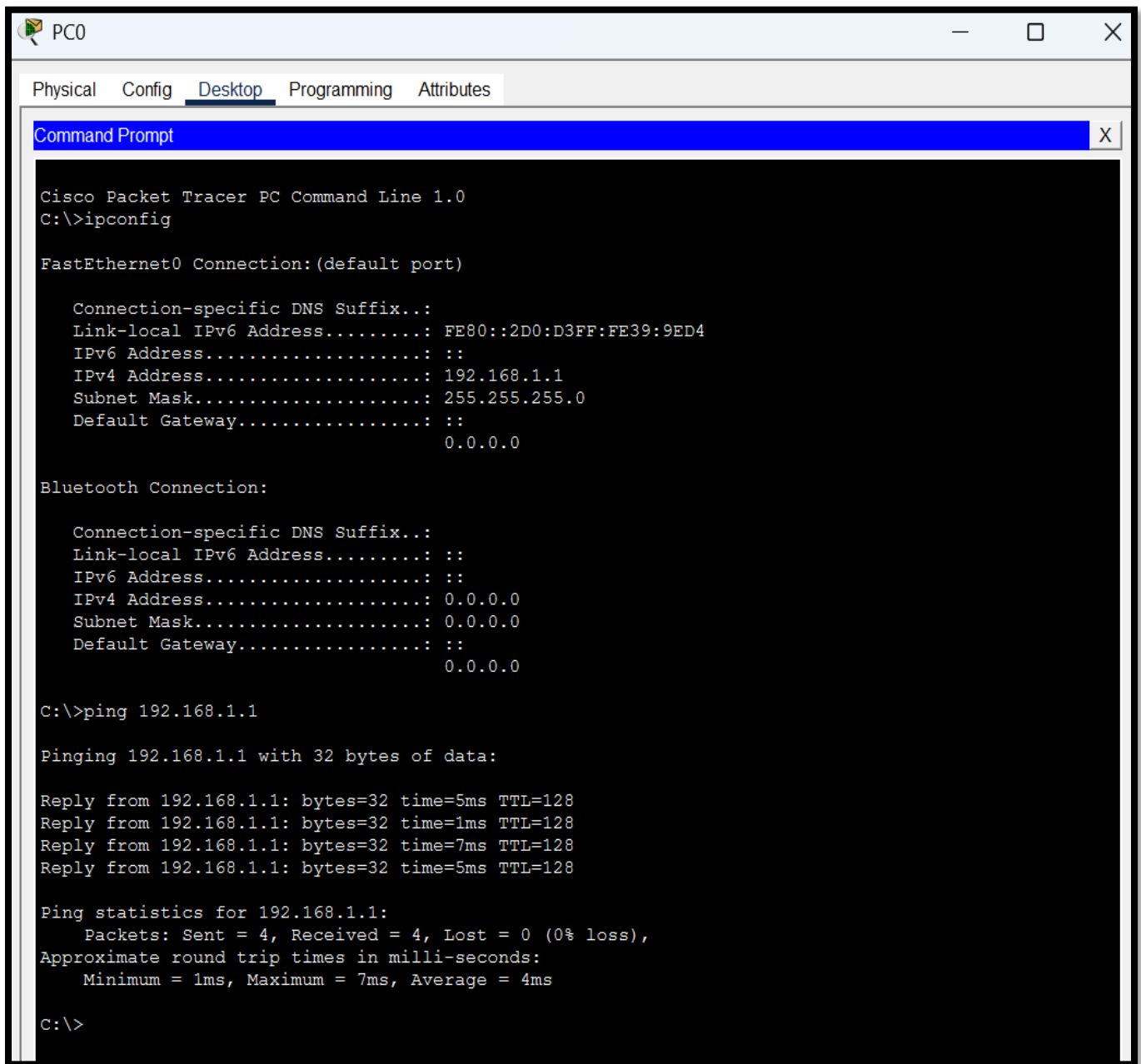
Step 03: 1. ipconfig

Displays the current IP address configuration of the PC.



Step 04: 2. *ping*

Tests connectivity between two network devices.



```
PC0
Physical Config Desktop Programming Attributes
Command Prompt X
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig

FastEthernet0 Connection: (default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::2D0:D3FF:FE39:9ED4
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 192.168.1.1
    Subnet Mask . . . . .: 255.255.255.0
    Default Gateway . . . . .: ::
                                   0.0.0.0

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: ::
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .: ::
                                   0.0.0.0

C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

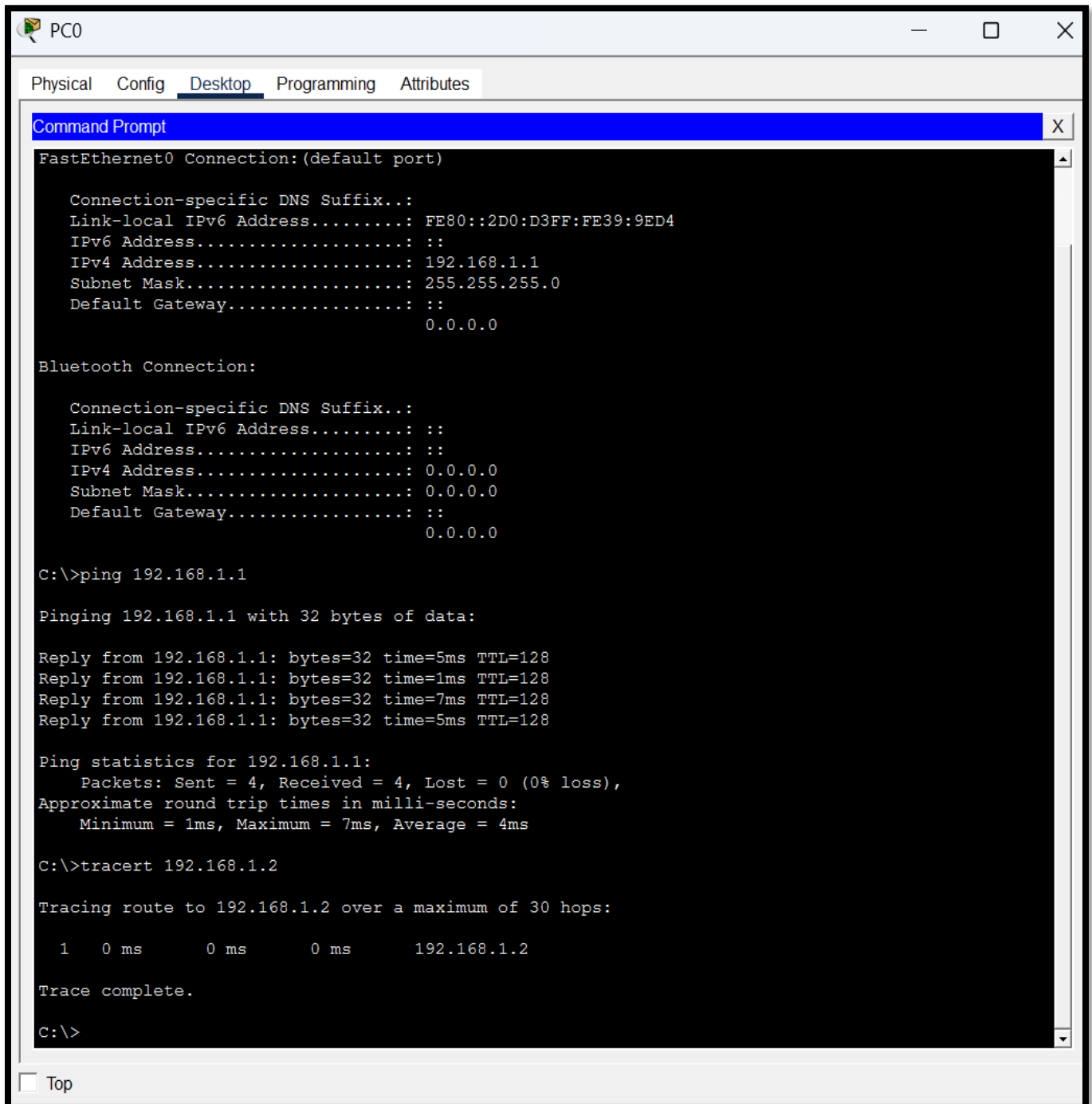
Reply from 192.168.1.1: bytes=32 time=5ms TTL=128
Reply from 192.168.1.1: bytes=32 time=1ms TTL=128
Reply from 192.168.1.1: bytes=32 time=7ms TTL=128
Reply from 192.168.1.1: bytes=32 time=5ms TTL=128

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 7ms, Average = 4ms

C:\>
```

Step 05: 3. *tracert*

Shows the path taken by packets to reach the destination.



The screenshot shows a window titled "PC0" with tabs for "Physical", "Config", "Desktop", "Programming", and "Attributes". The "Desktop" tab is active, displaying a "Command Prompt" window. The Command Prompt shows the configuration for "FastEthernet0" and "Bluetooth" connections, followed by a successful ping and a successful tracert to 192.168.1.2.

```
PC0
Physical Config Desktop Programming Attributes
Command Prompt
FastEthernet0 Connection: (default port)

Connection-specific DNS Suffix...:
Link-local IPv6 Address . . . . .: FE80::2D0:D3FF:FE39:9ED4
IPv6 Address . . . . .: ::
IPv4 Address . . . . .: 192.168.1.1
Subnet Mask . . . . .: 255.255.255.0
Default Gateway . . . . .: ::
                                0.0.0.0

Bluetooth Connection:

Connection-specific DNS Suffix...:
Link-local IPv6 Address . . . . .: ::
IPv6 Address . . . . .: ::
IPv4 Address . . . . .: 0.0.0.0
Subnet Mask . . . . .: 0.0.0.0
Default Gateway . . . . .: ::
                                0.0.0.0

C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time=5ms TTL=128
Reply from 192.168.1.1: bytes=32 time=1ms TTL=128
Reply from 192.168.1.1: bytes=32 time=7ms TTL=128
Reply from 192.168.1.1: bytes=32 time=5ms TTL=128

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 7ms, Average = 4ms

C:\>tracert 192.168.1.2

Tracing route to 192.168.1.2 over a maximum of 30 hops:

  1  0 ms    0 ms    0 ms    192.168.1.2

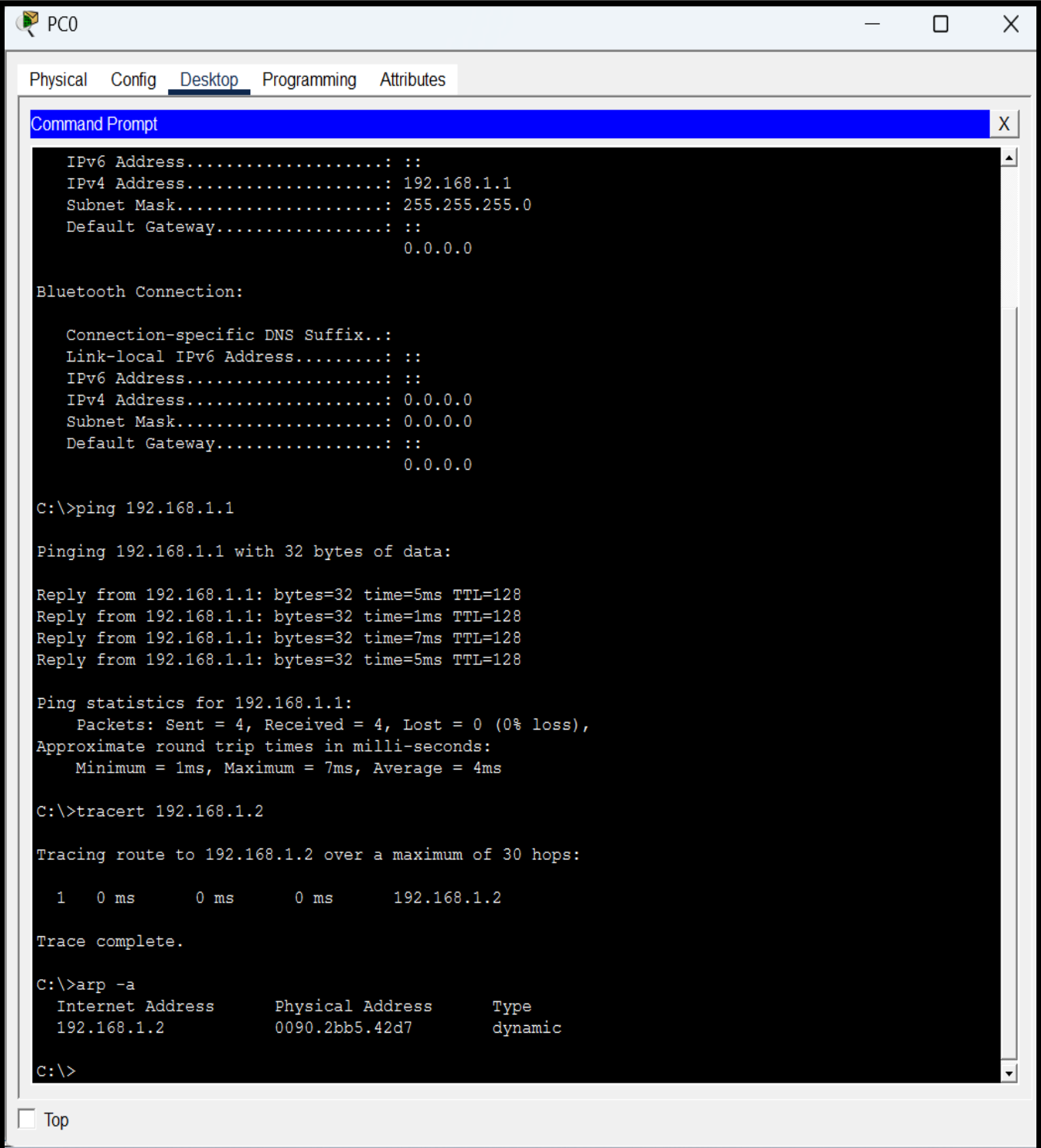
Trace complete.

C:\>
```

☐ Top

Step 06: 4. *arp -a*

Displays the ARP table (IP to MAC address mapping).



The screenshot shows a PC0 desktop environment with a taskbar at the top containing icons for Physical, Config, Desktop (selected), Programming, and Attributes. A Command Prompt window is open, displaying the following text:

```
IPv6 Address.....: ::
IPv4 Address.....: 192.168.1.1
Subnet Mask.....: 255.255.255.0
Default Gateway.....: ::
                        0.0.0.0

Bluetooth Connection:

Connection-specific DNS Suffix...:
Link-local IPv6 Address.....: ::
IPv6 Address.....: ::
IPv4 Address.....: 0.0.0.0
Subnet Mask.....: 0.0.0.0
Default Gateway.....: ::
                        0.0.0.0

C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time=5ms TTL=128
Reply from 192.168.1.1: bytes=32 time=1ms TTL=128
Reply from 192.168.1.1: bytes=32 time=7ms TTL=128
Reply from 192.168.1.1: bytes=32 time=5ms TTL=128

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 7ms, Average = 4ms

C:\>tracert 192.168.1.2

Tracing route to 192.168.1.2 over a maximum of 30 hops:

  1    0 ms    0 ms    0 ms    192.168.1.2

Trace complete.

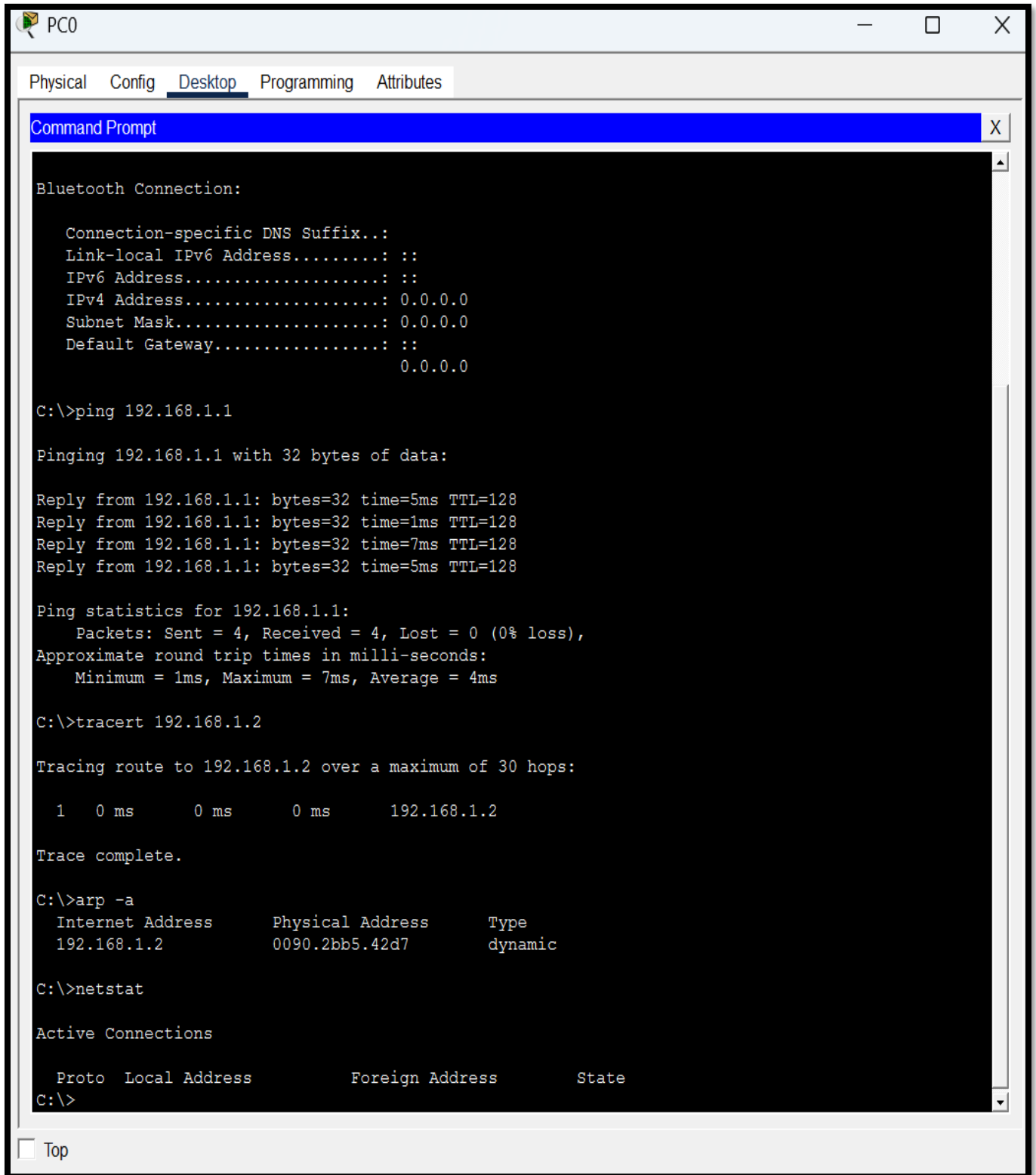
C:\>arp -a
Internet Address      Physical Address      Type
192.168.1.2           0090.2bb5.42d7        dynamic

C:\>
```

At the bottom left of the Command Prompt window, there is a checkbox labeled "Top" which is currently unchecked.

Step 07: 5. netstat

Displays active network connections and statistics.



The screenshot shows a PC0 window with a Command Prompt open. The window has tabs for Physical, Config, Desktop (selected), Programming, and Attributes. The Command Prompt title bar is blue and says "Command Prompt". The output of the commands is as follows:

```
Bluetooth Connection:

Connection-specific DNS Suffix...:
Link-local IPv6 Address.....: ::
IPv6 Address.....: ::
IPv4 Address.....: 0.0.0.0
Subnet Mask.....: 0.0.0.0
Default Gateway.....: ::
                        0.0.0.0

C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time=5ms TTL=128
Reply from 192.168.1.1: bytes=32 time=1ms TTL=128
Reply from 192.168.1.1: bytes=32 time=7ms TTL=128
Reply from 192.168.1.1: bytes=32 time=5ms TTL=128

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 7ms, Average = 4ms

C:\>tracert 192.168.1.2

Tracing route to 192.168.1.2 over a maximum of 30 hops:

  1  0 ms      0 ms      0 ms      192.168.1.2

Trace complete.

C:\>arp -a
Internet Address      Physical Address      Type
192.168.1.2           0090.2bb5.42d7        dynamic

C:\>netstat

Active Connections

   Proto  Local Address          Foreign Address         State
C:\>
```

At the bottom left of the window, there is a checkbox labeled "Top" which is currently unchecked.